

Model: $Y = f(X) + \epsilon_i$ & $E(\epsilon_i) = 0$

goal of minimize test MSE

[1.1] prove how $f_{opt} = E(Y|X)$ minimizes MSPE : $E[(Y - f(x))^2]$

Use The Law of Total Expectations : $E(X) = E(E(Y|X))$

$\therefore E[(Y - f(x))^2] = E[E[(Y - f(x))^2 | X]]$
 v.e.g. avg. over Y given X than avg over X

If we condition on $X=x$ (fixed value) $\sim f(x)$ is then a constant $\mu(x)$ is defined as $E(Y|X=x)$

$\rightarrow$ we can write $Y - f(x) = Y - \underbrace{\mu(x)}_{\text{cancel each other out}} + \underbrace{\mu(x) - f(x)}_{\text{constant given } X=x} = (Y - \mu(x)) + (\mu(x) - f(x))$

$\rightarrow [Y - f(x)]^2 = [Y - \mu(x)]^2 + 2[Y - \mu(x)][\mu(x) - f(x)] + [\mu(x) - f(x)]^2$

$\therefore E[Y - f(x)^2 | X=x] = E[\underbrace{[Y - \mu(x)]^2}_{\text{det cond. variance}} | X=x] + 2[Y - \mu(x)]E[\underbrace{Y - \mu(x)}_{\text{constant}} | X=x] + E[\underbrace{[\mu(x) - f(x)]^2}_{\text{property expected value e.g. } E[ax+b] = aE[X] + b} | X=x]$

$= \text{Var}(Y|X=x) + 2[Y - \mu(x)]\underbrace{[E[Y|X=x] - \mu(x)]}_{\text{eval to 0}} + [\mu(x) - f(x)]^2$
 $\uparrow$ a # since $X=x$

$= \underbrace{\text{Var}(Y|X=x)}_{\text{independent of } f \text{ irreducible error}} + \underbrace{[\mu(x) - f(x)]^2}_{\text{dependent on } f \text{ reducible}}$

Therefore Minimization of function relies only on the second term:

$\rightarrow \frac{\partial}{\partial f(x)} ([\mu(x) - f(x)]^2) = -2[\mu(x) - f(x)]$ & set = 0 the function is minimized when $f(x) = \mu(x) := E(Y|X=x)$

Therefore $f_{opt}(X) = \mu(x)$ & plug into $E[(Y - f(x))^2] = E[E[(Y - f(x))^2 | X]]$

we see that $E[E[(Y - f_{opt}(X))^2 | X]] = \text{Var}(Y|X)$ & taking expectation over X we see

$\sim$ minimal overall error: $E[(Y - f_{opt}(x))^2] = E[\text{Var}(Y|X)]$ which shows $f_{opt} = E[Y|X]$

[1.2] we have $f(x)$ the true func. & a obs at x_0 defined by $y_0 = f(x_0) + \epsilon_0$

$\sim E(\epsilon_0) = 0$ & $\text{var}(\epsilon_0) = \sigma^2$ Also $\hat{f}(x)$ tries to predict $f(x)$

'random training data'

Randomness comes from ϵ_0 which is independent of $\hat{f}$ & randomness in $\hat{f}(x_0)$

since we are taking expectation over both errors at the point x_0 :

$y_0 - \hat{f}(x_0) = f(x_0) + \epsilon_0 - \hat{f}(x_0) = \underbrace{[f(x_0) - \hat{f}(x_0)]}_{\text{accounts for variation}} + \epsilon_0$

Therefore: $E[(y_0 - \hat{f}(x_0))^2] = E[(f(x_0) - \hat{f}(x_0) + \epsilon_0)^2] = E[(f(x_0) - \hat{f}(x_0))^2] + 2E[(f(x_0) - \hat{f}(x_0))\epsilon_0] + E[\epsilon_0^2]$
 $= E[(f(x_0) - \hat{f}(x_0))^2] + \sigma^2$
 $\uparrow$ var(ϵ)
 $\uparrow$ multiply by $E[\epsilon_0]$ which is 0 since error has mean of 0

Decompose the first term: $A = \hat{f}(x_0)$ & $b = f(x_0)$ & det of variance $\text{var}(A) = E[A^2] - (E[A])^2$

$\rightarrow \text{Var}(A) = E[(A - b)^2] - (E[A - b])^2 \rightarrow E[(A - b)^2] = \text{Var}(A) + (E[A] - b)^2$

$\therefore E[f(x_0) - \hat{f}(x_0)]^2 = \text{var}(\hat{f}(x_0)) + (E[\hat{f}(x_0)] - f(x_0))^2$

NOTE, by def. Bias $(\hat{f}(x_0)) = E[\hat{f}(x_0)] - f(x_0)$

or σ^2

Therefore Plugging Things in: $E[(y_0 - \hat{f}(x_0))^2] = \underbrace{\text{var}(\hat{f}(x_0))}_{\text{reducible}} + [\text{Bias}(\hat{f}(x_0))]^2 + \underbrace{\text{var}(\epsilon)}_{\text{irreducible}}$